

TSO – Time-Sharing Option

Introduction





WHAT IS TIME SHARING?

Time-sharing enables more than one user of a computer system to access the system concurrently without interfering with each other. In the early days of mainframes, the systems could only execute one task at a time.

Since human interactions with the system take far more time than internal computing operations, when the system allows users to take turns, each user may feel as if they have the whole system to themselves – at least, most of the time.

Computers spend far more time waiting for us to decide what to do next than we spend waiting for them to complete whatever internal processing they are performing for us.



THE TIME-SHARING OPTION



When the IBM System/360 was introduced (1964), the standard input device was a punched card reader like the museum piece shown at left. The standard input stream was, and still is, the SYSIN data set. To this day, SYSIN contains 80-character text records that duplicate the layout of 80-column punched cards.

Although these were batch processing systems, it wasn't long before customers demanded interactive support. In 1971, when IBM released the Time-Sharing Option (TSO), they still believed most customers would be satisfied running punched cards in batch mode. It was offered as an add-on product.

TSO didn't become a part of the base operating system configuration until the advent of z/OS in 2001. The product name still includes the word, "option."

BLOCK-MODE TERMINALS AND AID KEYS

We mentioned 3270 terminal emulation when we were setting up the environment for the bootcamp. This type of device was state-of-the-art at the time TSO came out. It became the standard terminal for the IBM mainframe line.

This is a block-mode terminal, as opposed to character-mode devices such as the DEC VT-100. In a nutshell, that means that only certain keys caused data to be transmitted to the host. The other keys updated the contents of a buffer located in the terminal. The keys that sent data to the host are called *attention identifier* keys, or AID keys.

The AID keys are Enter, Clear, System Request, Program Attention (PA) 1 and 2, and Program Function 1 through 12 (up to 24 on later models). Several of these don't exist on standard keyboards today. Your terminal emulator software will offer a way to simulate these keys when you're using TSO.



TERMINAL EMULATOR MENU

File Options Keypad

IBM z/OS Standard Development/Test Image

Built on: 2024-03-21-16:35

System: VS01

Workstation IP: 71.226.56.53

Date: 01/26/26

Time: 10:27:34

To login to an Application type one of:

--> LOGON XXX (XXX is Vtam APPL)

--> TSO userid

--> CICSTS61

--> IMS15APL

--> CNM01

ENTER CHOICE ==> █

The special keys on 3270 terminals are not present on modern keyboards.

The Brain-in-a-Jar thinks it's talking to a real 3270 terminal, so we need a way to transmit the special key codes to it.

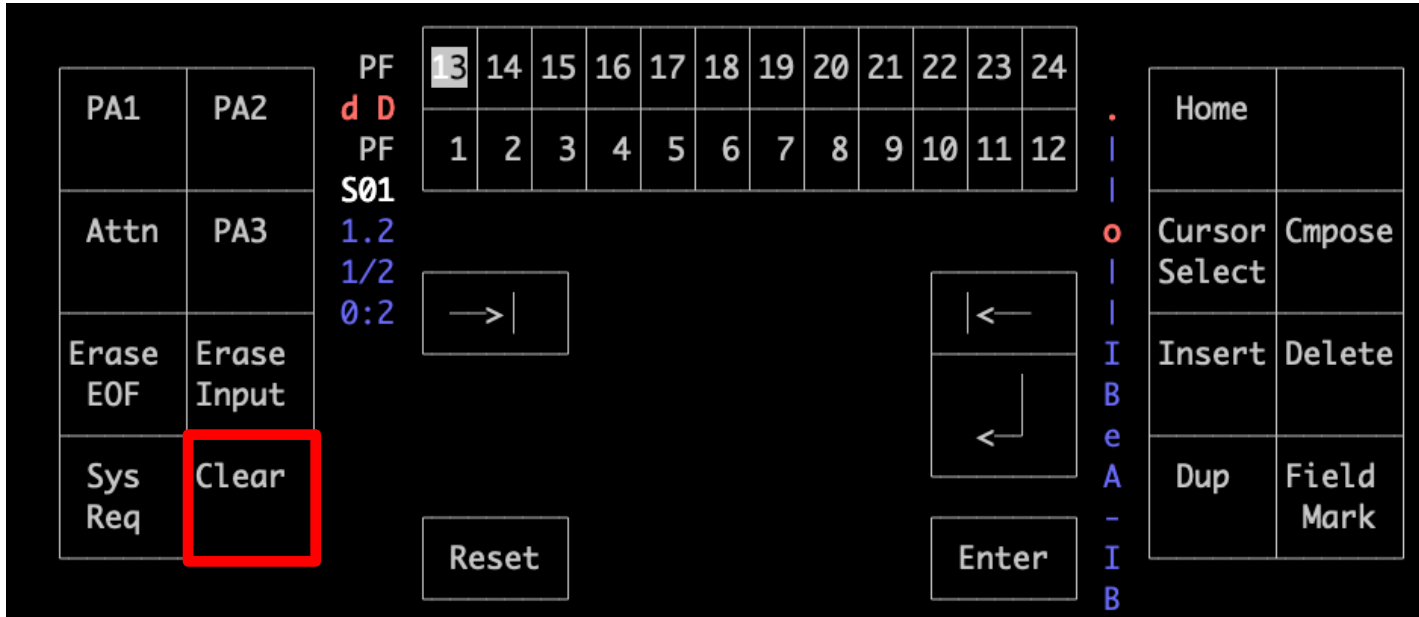
Terminal emulation software provides a way to simulate these keys. Each product has its own implementation of this.

The product I use is c3270, a command-line wrapper for the open source emulator, x3270. It provides a menu bar at the top of the screen, as shown at left.

When I need one of the special 3270 keys, I choose the Keypad option on that menu.

(_____) ||-|| |□□□□|
||||||| ||-|| |□□□□|

TERMINAL EMULATOR KEYPAD

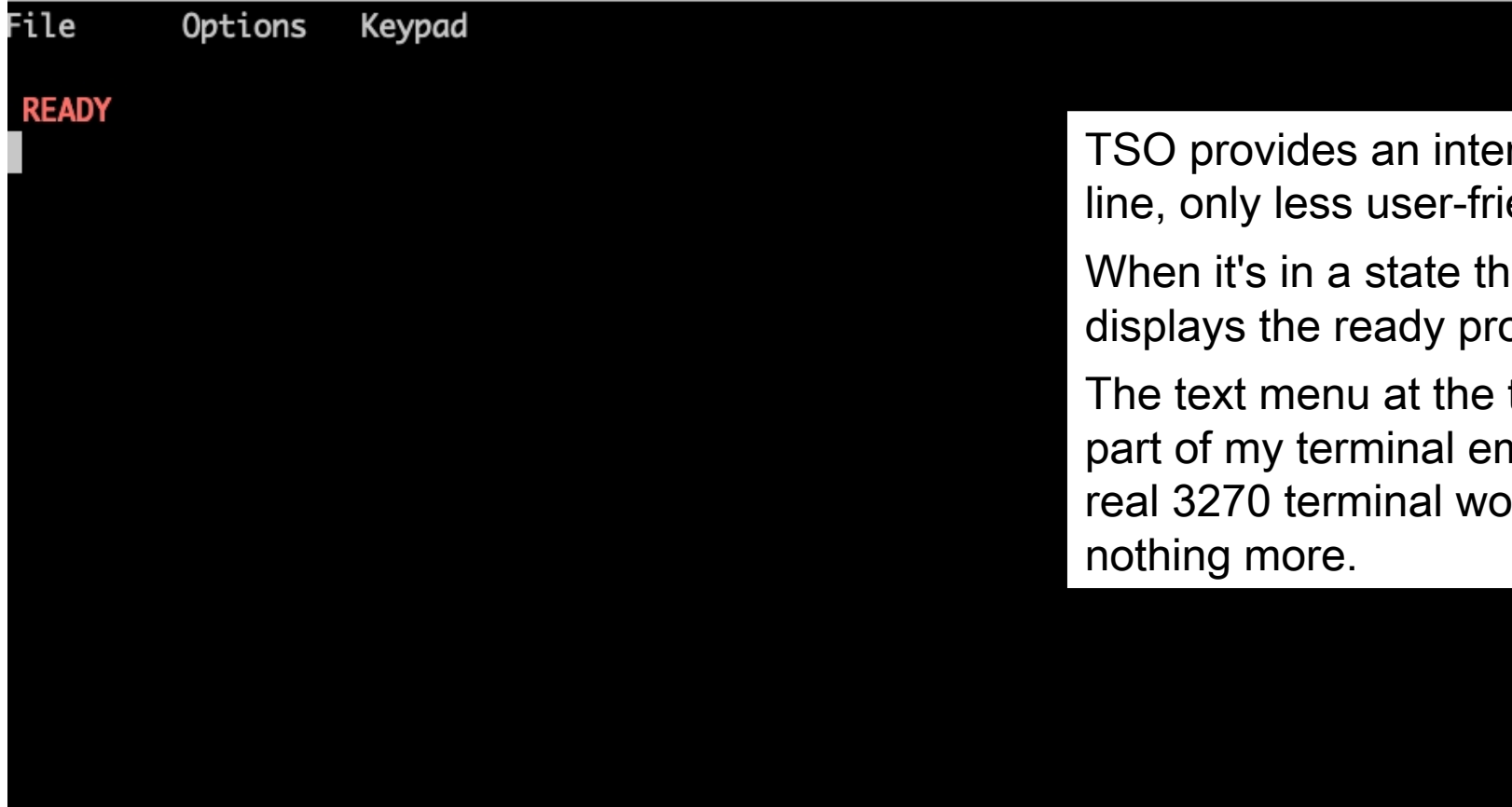


The Keypad option overlays a virtual keypad on the screen. A key can be selected with a mouse click. The emulator transmits the corresponding key code to the mainframe.

For instance, if I need to simulate pressing the Clear key on a 3270 keyboard, I click on the box labeled "Clear" on the virtual keypad.

This is only an example. The terminal emulator you're using may implement this functionality differently, but the special keys are available somewhere in the product's user interface.

THE TSO READY PROMPT



TSO provides an interface similar to a command line, only less user-friendly.

When it's in a state that can accept input, TSO displays the ready prompt.

The text menu at the top of the screen shot is part of my terminal emulator display. TSO on a real 3270 terminal would just say "ready" and nothing more.

THE TSO READY PROMPT

```
File      Options  Keypad

READY
time
IKJ56650I TIME-05:09:12 PM. CPU-00:00:00 SERVICE-1814 SESSION-00:03:23 FEBRUARY
16,2024
READY
█
```

We can enter a TSO command now. The command, TIME, will display the system time. When the command finishes, TSO displays the ready prompt again.

THE TSO READY PROMPT

```
File      Options  Keypad

READY
time
IKJ56650I TIME-05:09:12 PM. CPU-00:00:00 SERVICE-1814 SESSION-00:03:23 FEBRUARY
16,2024
READY
profile
IKJ56688I CHAR(0)  LINE(0)   PROMPT  INTERCOM  NOPAUSE MSGID  NOMODE  WTPMS
G  NORECOVER PREFIX(MATEVA)  PLANGUAGE(ENU) SLANGUAGE(ENU) VARSTORAGE(LOW)
IKJ56689I DEFAULT LINE/CHARACTER DELETE CHARACTERS IN EFFECT FOR THIS TERMINAL
READY
█
```

The PROFILE command will display the current settings of my TSO user profile.

THE TSO READY PROMPT

```
File      Options  Keypad

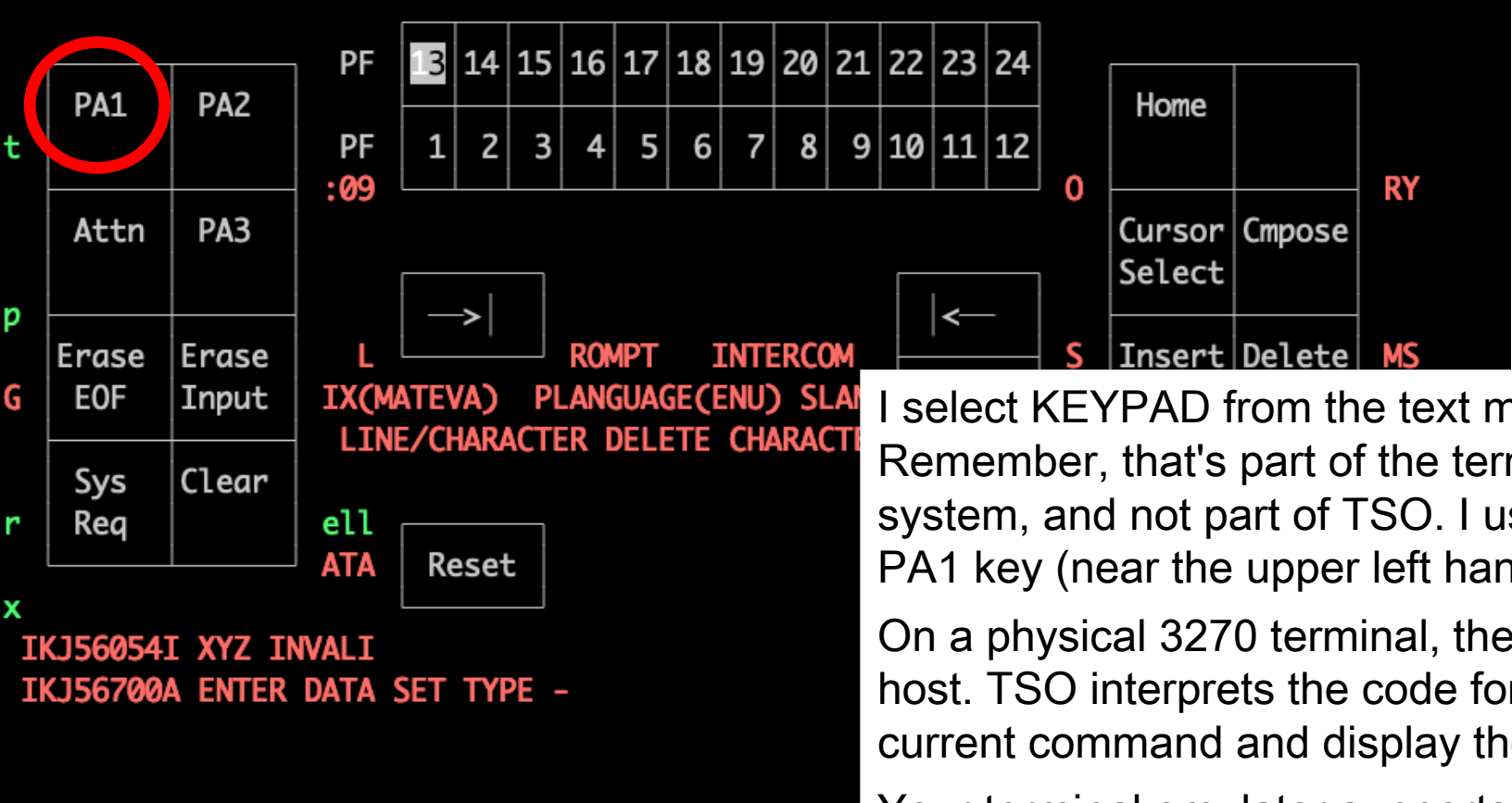
READY
time
IKJ56650I TIME-05:09:12 PM. CPU-00:00:00 SERVICE-1814 SESSION-00:03:23 FEBRUARY
16,2024
READY
profile
IKJ56688I CHAR(0)  LINE(0)   PROMPT  INTERCOM  NOPAUSE MSGID   NOMODE  WTPMS
G  NORECOVER PREFIX(MATEVA)  PLANGUAGE(ENU) SLANGUAGE(ENU) VARSTORAGE(LOW)
IKJ56689I DEFAULT LINE/CHARACTER DELETE CHARACTERS IN EFFECT FOR THIS TERMINAL
READY
run cobol.source(hello)
IKJ56700A ENTER DATA SET TYPE -
xyz
IKJ56054I XYZ INVALID DATA TYPE
IKJ56700A ENTER DATA SET TYPE -
█
```

Now let's make a mistake and see how to interrupt the TSO command in progress.

Here I've entered the RUN command and specified a data set name. TSO prompts for the data set type, and I respond with an invalid value.

I don't remember the valid values, so I want to blow out of this command and try it again later.

THE TSO READY PROMPT



I select KEYPAD from the text menu at the top of the display. Remember, that's part of the terminal emulator on my local system, and not part of TSO. I use the keypad to "press" the PA1 key (near the upper left hand corner).

On a physical 3270 terminal, the PA1 key sends a code to the host. TSO interprets the code for PA1 to mean "interrupt the current command and display the ready prompt".

Your terminal emulator supports simulating a PA1 keypress, even if it doesn't look identical to this example.

In any case, when you get stuck in TSO, PA1 is the way out.

THE TSO READY PROMPT

```
File      Options  Keypad

READY
time
IKJ56650I TIME-05:09:12 PM. CPU-00:00:00 SERVICE-1814 SESSION-00:03:23 FEBRUARY
16,2024
READY
profile
IKJ56688I CHAR(0)  LINE(0)   PROMPT  INTERCOM  NOPAUSE MSGID  NOMODE  WTPMS
G  NORECOVER PREFIX(MATEVA)  PLANGUAGE(ENU) SLANGUAGE(ENU) VARSTORAGE(LOW)
IKJ56689I DEFAULT LINE/CHARACTER DELETE CHARACTERS IN EFFECT FOR THIS TERMINAL
READY
run cobol.source(hello)
IKJ56700A ENTER DATA SET TYPE -
xyz
IKJ56054I XYZ INVALID DATA TYPE
IKJ56700A ENTER DATA SET TYPE -
|
READY
logoff
```

To log off, type LOGOFF and press enter.

ISPF – INTERACTIVE SYSTEM PRODUCTIVITY FACILITY

We don't spend much time using TSO the way we've been demonstrating it. We won't dwell on all the available commands.

Instead, we use a full-screen environment that wraps the TSO commands plus a lot more – ISPF.

ISPF is a huge improvement over basic TSO. It's tremendously customizable and extensible, but still less polished than modern-day IDEs and smart editors.

ISPF is the one of the three pure-text editors and development environments that came from the era before graphical user interfaces were common and have stood the test of time.

You may already use one or both of the others – EMACS and VIM. Those are children of Unix. ISPF is just for z/OS, but in its heyday at least as many developers used ISPF daily as those who used EMACS and VIM.

```
Menu  Utilities  Compilers  Options  Status  Help
-----
ISPF Primary Option Menu

Option ==> |

0 Settings      Terminal and user parameters      User ID . : IBMUSER
1 View          Display source data or listings   Time. . . : 10:34
2 Edit          Create or change source data      Terminal. : 3278
3 Utilities     Perform utility functions        Screen. . : 1
4 Foreground    Interactive language processing  Language. : ENGLISH
5 Batch         Submit job for language processing Appl ID . : ISR
6 Command       Enter TSO commands               TSO logon : DB13PROC
7 Dialog Test   Perform dialog testing           TSO prefix: IBMUSER
9 IBM Products  IBM program development products  System ID : VS01
10 SCLM         SW Configuration Library Manager  MVS acct. : ACCT001
11 Workplace    ISPF Object/Action Workplace     Release . : ISPF 8.1
12 z/OS System  z/OS system programmer applications
13 z/OS User    z/OS user applications

Enter X to Terminate using log/list defaults
```

HOW MUCH DO YOU NEED TO KNOW ABOUT ISPF?

As a platform engineer who performs most of the work using off-platform tools like Ansible, you don't have to become an expert with TSO and ISPF.

Because Ansible is a leaky abstraction over z/OS system commands, you will have to sign on directly to a z/OS instance occasionally to verify the effects of your Ansible playbooks or to troubleshoot issues.

For learning purposes, we'll spend some time working directly on the z/OS instance. There isn't much use trying to automate things that you don't understand at all.

For your role, you'll need enough knowledge of TSO and ISPF to

- find data sets and view their attributes;
- create and delete data sets;
- perform basic editing operations on data sets;
- submit jobs;
- view and delete the output from jobs (SYSOUT, SYSPRINT); and
- position the cursor to enable your 3270 emulator to transfer files.

ISPF – INTERACTIVE SYSTEM PRODUCTIVITY FACILITY

```
User ID . : IBMUSER
Time. . . : 10:34
Terminal. : 3278
Screen. . : 1
Language. : ENGLISH
Appl ID . : ISR
TSO logon : DB13PROC
TSO prefix: IBMUSER
System ID : VS01
MVS acct. : ACCT001
Release . : ISPF 8.1
```

Let's explore the initial ISPF display. We'll start by zooming in on the status information displayed on the right side.

What can we teach ourselves by examining these values?

Here's a small challenge: Find the answers to the following questions in the screenshot.

- 1) What's the system ID of the z/OS instance we're connected to?
- 2) What release of ISPF are we using?
- 3) Which particular 3270 model is our terminal emulator pretending to be?
- 4) Is it possible to have more than one active screen under ISPF?
- 5) Is it possible for ISPF to display information in languages other than English?

ISPF – INTERACTIVE SYSTEM PRODUCTIVITY FACILITY

The main part of the display shows the Primary Option Menu. We choose an option from the menu by entering its code in the Option field and pressing Enter.

Before we do that, let's zoom in on the menu and check it out.

Everything on this menu is configurable. However, nearly all z/OS installations leave options 0 through 6 alone. The rest might be different, depending on how the particular z/OS instance is configured and what optional products may be installed on it.

The menu may look different based on the user's authorization to access various system facilities. For instance, an application developer would not see option 12 on this menu.

ISPF Primary Option Menu		
Option ==>		
0	Settings	Terminal and user parameters
1	View	Display source data or listings
2	Edit	Create or change source data
3	Utilities	Perform utility functions
4	Foreground	Interactive language processing
5	Batch	Submit job for language processing
6	Command	Enter TSO commands
7	Dialog Test	Perform dialog testing
9	IBM Products	IBM program development products
10	SCLM	SW Configuration Library Manager
11	Workplace	ISPF Object/Action Workplace
12	z/OS System	z/OS system programmer applications
13	z/OS User	z/OS user applications
Enter X to Terminate using log/list defaults		

ISPF – INTERACTIVE SYSTEM PRODUCTIVITY FACILITY

F1=Help F2=Split F3=Exit F7=Backward F8=Forward F9=Swap
F10=Actions F12=Cancel

Along the bottom of the display, there are some hints about what the Program Function (PF) keys do. This is all configurable, but nearly all installations leave these settings alone. These functions are consistent throughout ISPF panels:

PF1 = Help

PF8 = Scroll forward or show the next "thing"

PF7 = Scroll backward or show the previous "thing"

PF3 = Save changes and exit the current panel

PF12 = Cancel changes and exit the current panel

ISPF – INTERACTIVE SYSTEM PRODUCTIVITY FACILITY

Along the top of the display, there's a text menu bar. This menu is part of ISPF, not part of the terminal emulator.

The items on this menu correspond to options you can select from the Primary Option Menu or by pressing a PF key. The menu is provided in case you re-map the standard function keys to other functions. You still have a way to reach any option you need.

If you're using a terminal emulator on a GUI desktop, you can move the cursor by clicking somewhere in the display window, but ISPF does not respond to pointer gestures. It's a pure text environment. So normally, you'd move the cursor around with the tab key and arrow keys.

```
Menu Utilities Compilers Options Status Help
ISPF Primary Option Menu
Option ==>
0 Settings      Terminal and user parameters      User ID . : IBMUSER
1 View          Display source data or listings    Time. . . : 10:34
2 Edit          Create or change source data       Terminal. : 3278
3 Utilities     Perform utility functions         Screen. . : 1
4 Foreground    Interactive language processing   Language. : ENGLISH
5 Batch         Submit job for language processing  Appl ID . : ISR
6 Command       Enter TSO commands                TSO logon : DB13PROC
7 Dialog Test   Perform dialog testing            TSO prefix: IBMUSER
9 IBM Products  IBM program development products   System ID : VS01
10 SCLM         SW Configuration Library Manager   MVS acct. : ACCT001
11 Workplace    ISPF Object/Action Workplace       Release . : ISPF 8.1
12 z/OS System  z/OS system programmer applications
13 z/OS User    z/OS user applications

Enter X to Terminate using log/list defaults
```


ISPF – INTERACTIVE SYSTEM PRODUCTIVITY FACILITY

```
Menu Utilities Compilers Options Status Help
┌ 1. Settings
├ 2. View
├ 3. Edit
├ 4. ISPF Command Shell
├ 5. Dialog Test...
├ 6. Other IBM Products...
├ 7. SCLM
├ 8. ISPF Workplace
├ 9. Status Area...
└ 10. Exit

Primary Option Menu

er parameters      User ID . : IBMUSER
data or listings   Time. . . : 10:41
e source data      Terminal. : 3278
functions          Screen. . : 1
guage processing   Language. : ENGLISH
language processing Appl ID . : ISR
nds               TSO logon : DB13PROC
testing           TSO prefix: IBMUSER
                  System ID : VS01
9 IBM Products    IBM program development products
10 SCLM           SW Configuration Library Manager
11 Workplace     ISPF Object/Action Workplace
12 z/OS System    z/OS system programmer applications
13 z/OS User      z/OS user applications

Enter X to Terminate using log/list defaults
```

Positioning the cursor anywhere on the word, Menu, and pressing Enter causes this pop-up menu to appear. It contains several of the same options as we find on the Primary Option Menu. Those items activate the same functions as the primary options.

ISPF – INTERACTIVE SYSTEM PRODUCTIVITY FACILITY

Positioning the cursor anywhere on the word, Utilities, and pressing Enter causes this pop-up menu to appear. It contains several of the same options we can reach through Primary Menu Option 3, Utilities.

```
Menu  Utilities  Compilers  Options  Status  Help
-----
Optio 1. Library      Primary Option Menu
       2. Data set
       3. Move/Copy
0 Se   4. Data Set List  r parameters      User ID . : IBMUSER
1 Vi   5. Reset Statistics  ata or listings   Time. . . : 10:41
2 Ed   6. Hardcopy        source data       Terminal. : 3278
3 Ut   *. Reserved       functions         Screen. . : 1
4 Fo   8. Outlist         uage processing   Language. : ENGLISH
5 Ba   9. Commands...    anguage processing Appl ID . : ISR
6 Co   *0. Reserved      ds               TSO logon : DB13PROC
7 Di   11. Format         esting           TSO prefix: IBMUSER
9 IB   12. SuperC         lopment products System ID : VS01
10 SC  13. SuperCE        Library Manager   MVS acct. : ACCT001
11 Wo  14. Search-For     on Workplace      Release . : ISPF 8.1
12 z/  15. Search-ForE    rammer applications
13 z/  16. Table Utility   ations
       17. Directory List
                                og/list defaults
```

ISPF – INTERACTIVE SYSTEM PRODUCTIVITY FACILITY

```
Menu Utilities Compilers Options Status Help
Option ==>
0 Settings
1 View
2 Edit      Create or change source data
3 Utilities Perform utility functions
4 Foreground Interactive language processing
5 Batch     Submit job for language processing
6 Command   Enter TSO commands
7 Dialog Test Perform dialog testing
9 IBM Products IBM program development products
10 SCLM      SW Configuration Library Manager
11 Workplace ISPF Object/Action Workplace
12 z/OS System z/OS system programmer applications
13 z/OS User  z/OS user applications

1. Foreground Compilers
2. Background Compilers
3. ISPPREP Panel Utility...
4. DTL Compiler...

User ID . : IBMUSER
Time. . . : 10:41
Terminal. : 3278
Screen. . : 1
Language. : ENGLISH
Appl ID . : ISR
TSO logon : DB13PROC
TSO prefix: IBMUSER
System ID : VS01
MVS acct. : ACCT001
Release . : ISPF 8.1

Enter X to Terminate using log/list defaults
```

Positioning the cursor anywhere on the word, Compilers, and pressing Enter causes this pop-up menu to appear.

ISPF – INTERACTIVE SYSTEM PRODUCTIVITY FACILITY

Positioning the cursor anywhere on the word, Options, and pressing Enter causes this pop-up menu to appear.

```
Menu Utilities Compilers Options Status Help
Option ==>
0 Settings      Terminal a
1 View          Display so
2 Edit          Create or
3 Utilities     Perform ut
4 Foreground    Interactive language processing
5 Batch         Submit job for language processing
6 Command       Enter TSO commands
7 Dialog Test   Perform dialog testing
9 IBM Products  IBM program development products
10 SCLM         SW Configuration Library Manager
11 Workplace    ISPF Object/Action Workplace
12 z/OS System  z/OS system programmer applications
13 z/OS User    z/OS user applications

1. General Settings
2. CUA Attributes...
3. Keylists...
4. Point-and-Shoot...
5. Colors...
6. Dialog Test appl ID...

ID . : IBMUSER
. . . : 10:41
inal. : 3278
en. . : 1
Language. : ENGLISH
Appl ID . : ISR
TSO logon : DB13PROC
TSO prefix: IBMUSER
System ID : VS01
MVS acct. : ACCT001
Release . : ISPF 8.1

Enter X to Terminate using log/list defaults
```

ISPF – INTERACTIVE SYSTEM PRODUCTIVITY FACILITY

```
Menu  Utilities  Compilers  Options  Status  Help
-----
Option ==>      ISPF Prim
0 Settings      Terminal and user p
1 View          Display source data
2 Edit          Create or change so
3 Utilities      Perform utility fun
4 Foreground     Interactive language processing
5 Batch         Submit job for language processing
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7 Dialog Test   Perform dialog testing
9 IBM Products  IBM program development products
10 SCLM         SW Configuration Library Manager
11 Workplace    ISPF Object/Action Workplace
12 z/OS System  z/OS system programmer applications
13 z/OS User    z/OS user applications

    *. Session
    2. Function keys
    3. Calendar
    4. User status
    5. User point and shoot
    6. None

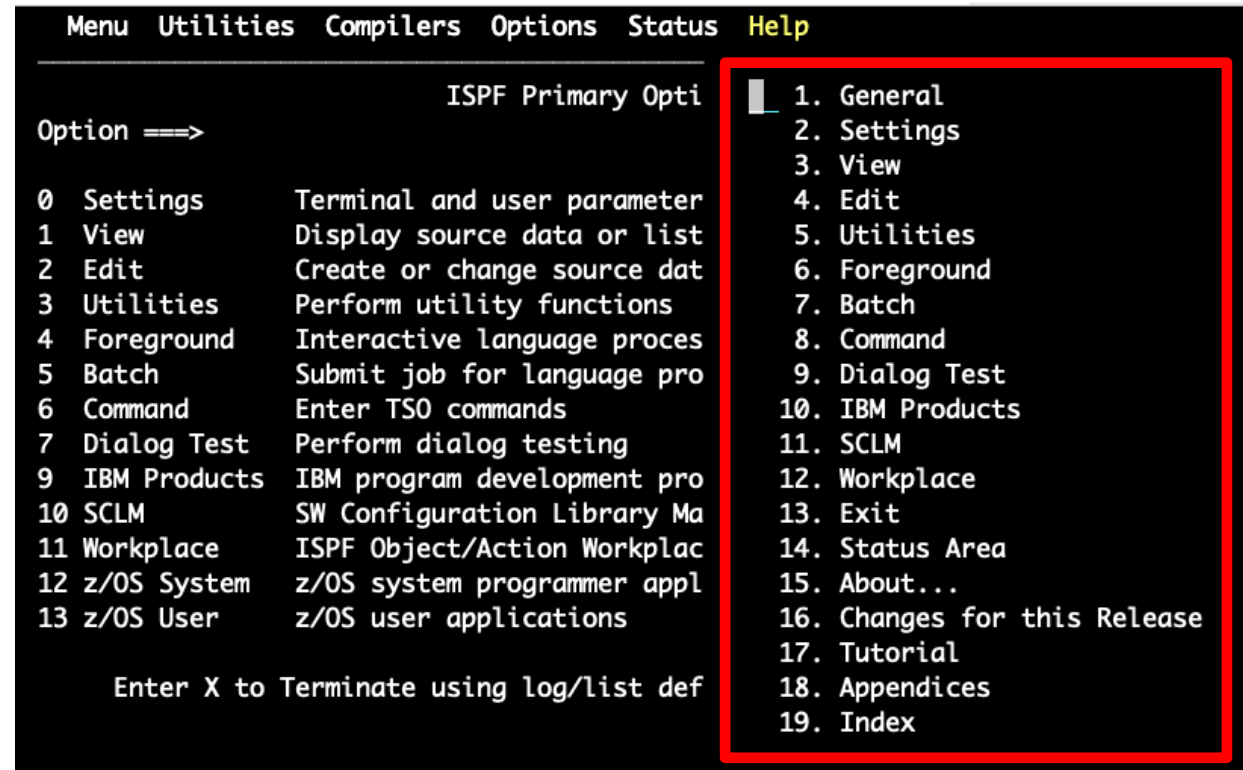
Language. : ENGLISH
Appl ID . : ISR
TSO logon : DB13PROC
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Release . : ISPF 8.1

Enter X to Terminate using log/list defaults
```

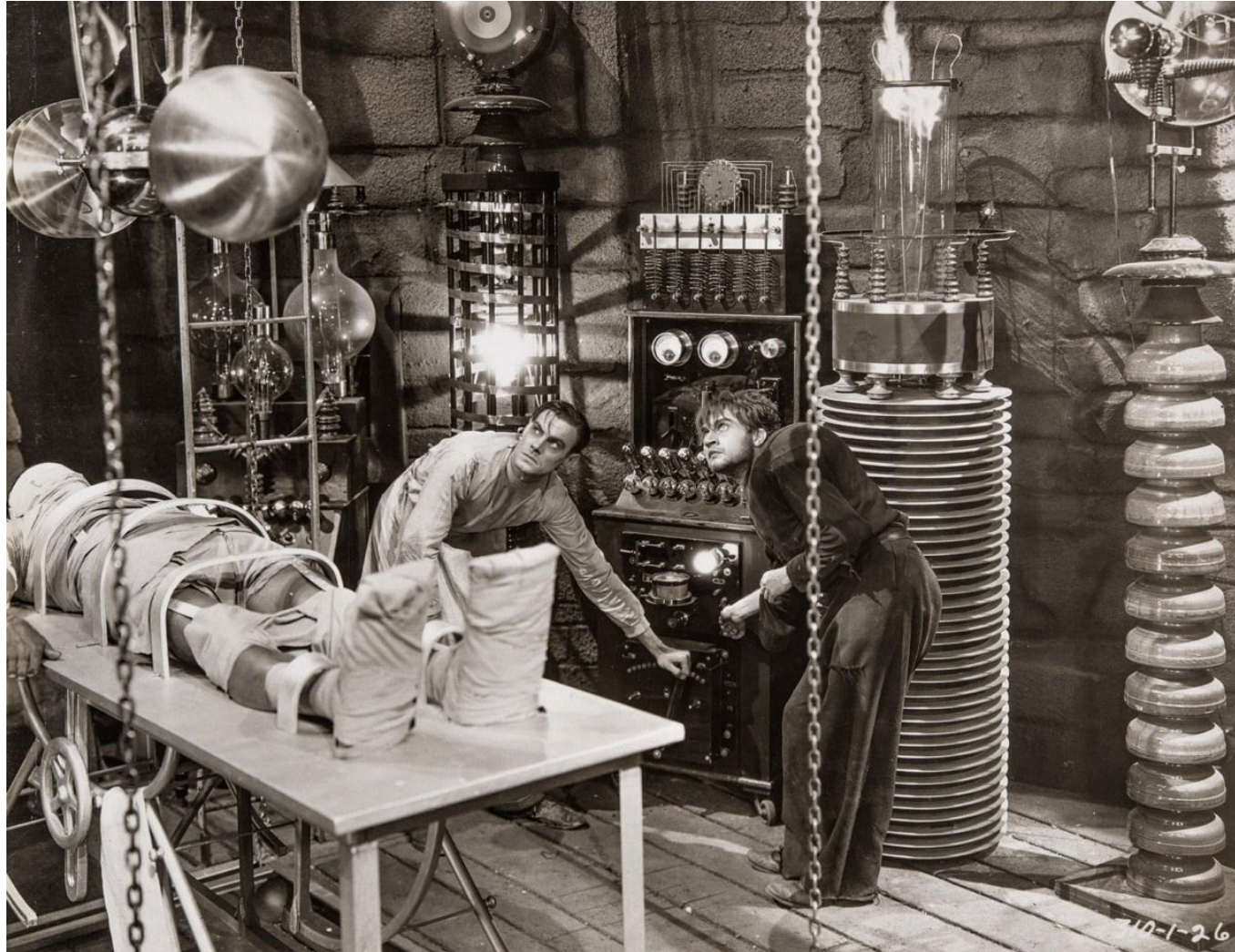
Positioning the cursor anywhere on the word, Status, and pressing Enter causes this pop-up menu to appear.

ISPF – INTERACTIVE SYSTEM PRODUCTIVITY FACILITY

Positioning the cursor anywhere on the word, Help, and pressing Enter causes this pop-up menu to appear. It offers a list of topics for which help is available.



LAB Z/OS 4: EXPLORING TSO AND ISPF



WHAT YOU KNOW AND WHAT YOU CAN DO

- You can log on and off TSO.
- You can start and exit ISPF.
- You can navigate around ISPF using the menu bar ("action bar") and the Primary Option Menu.
- You know which ISPF option to select if you want to customize the behavior of ISPF.

WHAT'S NEXT

- Creating a z/OS Data Set